

Channel representations of noisy coherent detection in continuous-variable quantum key distribution: Additive-noise versus attenuator

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We study the security of continuous-variable quantum key distribution (CV-QKD) under receiver asymmetry within an untrusted-detector model. We analyze additive-noise and attenuator channel representations of the same target POVM for each detection scheme. For homodyne and double-homodyne detection, we derive the corresponding channel parameters and compare the resulting Eve–Bob Holevo information and asymptotic secret fraction.

For the parameter sets considered, receiver asymmetry reduces the asymptotic secret fraction for both effective channels. The channel giving the lower asymptotic secret fraction depends on the receiver parameters and the transmittance of the communication channel. For homodyne detection, the attenuator channel gives a lower asymptotic secret fraction at high communication-channel transmittance. Its secret fraction remains positive over a longer distance.

In the double-homodyne case, the channel parameters, Eve–Bob Holevo information, and asymptotic secret fraction depend on the squeezing parameter r that labels the reference Gaussian POVM. For the parameter sets considered, the Eve–Bob Holevo information as a function of r is concave for the additive-noise channel and convex for the attenuator channel. The choice of r and the effective channel therefore affect the resulting asymptotic secret fraction.

I. INTRODUCTION

Homodyne and double-homodyne detection are standard coherent detection schemes in quantum optics and continuous-variable quantum information [1, 2]. Here double-homodyne detection denotes the two-branch receiver that implements the conventional heterodyne measurement. In the strong local oscillator limit, an ideal homodyne receiver implements a projective measurement of one field quadrature. An ideal balanced double-homodyne receiver jointly estimates two conjugate quadratures through a coherent-state positive operator-valued measure (POVM), whose effects are proportional to coherent-state projectors [2, 3]. Both schemes implement Gaussian POVMs and give Gaussian outcome distributions for Gaussian input states. They are used in quantum state tomography [4, 5], quantum state preparation [6], and continuous-variable quantum key distribution (CV-QKD) [7–9].

Arthurs and Kelly derived an uncertainty relation for the output variances of unbiased simultaneous measurements of conjugate observables [10]. For two canonical quadratures, an explicit construction introduces an auxiliary mode prepared in a fixed state and measures two commuting quadrature combinations [11]. For a centered isotropic Gaussian state and a cost equal to the sum of the two mean-square errors, the coherent-state POVM with a vacuum auxiliary mode minimizes that cost [12]. Yuen and Chan later analyzed the mean-square noise spectra of optical homodyne and heterodyne detection, distinguishing signal quantum fluctuations from local-oscillator excess noise [13]. Holevo’s classification of Gaussian POVMs includes noiseless and noisy homodyne measurements and heterodyne measurements with vacuum or thermal noise [3].

Practical receivers may have detector quantum efficiencies below unity and beam-splitter imbalance. Within the Gaussian approximation to the photocount statistics in the strong local oscillator limit, these imperfections yield noisy Gaussian POVMs [14, 15]. These receiver imperfections enter CV-QKD parameter estimation and the calculation of the asymptotic secret fraction [16–18]. We use *receiver asymmetry* for the combined effect of beam-splitter imbalance and unequal detector quantum efficiencies. For each detection scheme, the receiver parameters determine the noisy Gaussian POVM to be reproduced, which we call the target POVM. A channel representation of the target POVM

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consists of an effective one-mode Gaussian channel $\mathcal{C}$ acting after the communication channel, a reference Gaussian POVM applied to its output, and an invertible deterministic rescaling of the measurement outcomes.

Our previous work derived the target POVMs for receiver asymmetry in homodyne and double-homodyne detection and constructed their additive-noise channel representations [19]. For double-homodyne detection, we used a family of rank-one reference Gaussian POVMs generated by displaced squeezed states. The squeezing parameter r labels this family; $r = 0$ gives the coherent-state POVM of conventional double-homodyne detection. Complete positivity of the additive-noise channel restricts r to an interval fixed by the receiver parameters. In that analysis, an eavesdropper (Eve) was assumed to purify the Alice–Bob state after the effective receiver channel, and the resulting Eve–Bob Holevo information depends on r . The same target POVM admits several such representations. Its outcome statistics alone do not determine the Alice–Bob state after the effective receiver channel or the corresponding Eve–Bob Holevo information.

Here we construct an attenuator channel representation of the target homodyne and double-homodyne POVMs and compare it with the additive-noise channel representation. We derive the corresponding channel parameters from the requirement that both representations reproduce the target POVM. We then compare the resulting Eve–Bob Holevo information and asymptotic secret fraction for the Gaussian-modulated coherent-state protocol with reverse reconciliation under collective Gaussian attacks [7, 20, 21].

We restrict the analysis to zero-displacement Gaussian channels with centered noise and use units in which the vacuum covariance matrix is $\mathbb{1}$. We specify both effective channels through their action on the first-moment vector $\mathbf{r}_G$ and covariance matrix Σ_G of a Gaussian state:

$$\mathbf{r}_G \mapsto X_C \mathbf{r}_G, \quad \Sigma_G \mapsto X_C \Sigma_G X_C^T + Y_C.$$

For measurement type $\gamma \in \{\text{H}, \text{DH}\}$, the additive-noise channel Φ_γ and the attenuator channel $\mathcal{E}_\gamma$ are specified by

$$\begin{aligned} X_\Phi^{(\gamma)} &= \mathbb{1}, & Y_\Phi^{(\gamma)} &= \Sigma_N^{(\gamma)} \succeq 0, \\ X_\mathcal{E}^{(\gamma)} &= \cos \theta_\gamma \mathbb{1}, & Y_\mathcal{E}^{(\gamma)} &= \sin^2 \theta_\gamma \Sigma_{\text{env}}^{(\gamma)}, \end{aligned}$$

Here H and DH label homodyne and double-homodyne detection. The matrix $\Sigma_N^{(\gamma)}$ is the receiver excess-noise covariance matrix added by Φ_γ . The quantity $\cos^2 \theta_\gamma$ is the attenuator transmissivity, with $0 \leq \theta_\gamma < \pi/2$, and $\Sigma_{\text{env}}^{(\gamma)}$ is the covariance matrix of a centered Gaussian state of the environmental input mode [2, 22]. For $\mathcal{C}_\gamma \in \{\Phi_\gamma, \mathcal{E}_\gamma\}$, the dual map $\mathcal{C}_\gamma^*$ is defined by $\text{Tr}[\mathcal{C}_\gamma(\hat{\rho})\hat{\Pi}] = \text{Tr}[\hat{\rho}\mathcal{C}_\gamma^*(\hat{\Pi})]$ for every state $\hat{\rho}$ and POVM effect $\hat{\Pi}$. We choose the channel parameters so that $\mathcal{C}_\gamma^*$, together with an invertible deterministic rescaling of the outcomes, maps the reference Gaussian POVM to the target POVM. The two channel representations therefore give the same receiver outcome statistics in the rescaled coordinates for every input state.

Let $\hat{\rho}_{\text{AB}}$ denote the state shared by Alice and Bob after the communication channel and before the effective receiver channel. Each effective receiver channel produces

$$\hat{\rho}_{\text{AB}}^{(\mathcal{C}_\gamma)} = (\mathcal{I}_A \otimes \mathcal{C}_\gamma)(\hat{\rho}_{\text{AB}}),$$

where $\mathcal{I}_A$ is the identity channel on Alice’s mode and $\mathcal{C}_\gamma$ acts on Bob’s mode. Because the outcome rescaling is invertible, both representations give the same Alice–Bob mutual information. In a trusted-detector model, Eve has no access to the environmental output of the effective receiver channel [23–26]. For a fixed joint state of Eve, Alice, and Bob before the effective receiver channel, the target POVM gives the same joint state of Eve and Bob’s classical outcome, and hence the same Eve–Bob Holevo information, for both channel representations.

In the untrusted-detector model adopted here, Eve holds a purification of $\hat{\rho}_{\text{AB}}^{(\mathcal{C}_\gamma)}$. The two effective receiver channels generally produce different states $\hat{\rho}_{\text{AB}}^{(\mathcal{C}_\gamma)}$ and can give different values of the Eve–Bob Holevo information and asymptotic secret fraction. Selecting an effective receiver channel therefore fixes an additional assumption in the security analysis.

Section II defines the target homodyne and double-homodyne POVMs. Section III constructs their additive-noise and attenuator channel representations. Section IV evaluates the Alice–Bob mutual information, Eve–Bob Holevo information, and asymptotic secret fraction for both effective channels. Section V discusses the results.

II. TARGET GAUSSIAN POVMS

We now specify the target homodyne and double-homodyne POVMs produced by receiver asymmetry, as shown in Fig. 1. Receiver asymmetry arises from beam-splitter imbalance and unequal detector quantum efficiencies. Reference [19] gives the detailed derivations used in this section.

For the homodyne receiver, $\hat{a}$ is the annihilation operator of the signal mode, α and α_L are the coherent amplitudes of the signal and local oscillator (LO), and C and S are real beam-splitter amplitude coefficients satisfying $C^2 + S^2 = 1$. The photodetectors have quantum efficiencies η_1 and η_2 , and $\mu = m_1 - m_2$ is their photon-count difference.

The Q symbol of the target homodyne POVM (see Fig. 1a) is

$$P_G^{(H)}(x) = \frac{1}{\sqrt{2\pi\sigma_x}} \exp\left[-\frac{(x - \langle\hat{x}_\phi\rangle)^2}{2\sigma_x}\right], \quad (1)$$

$$\langle\hat{x}_\phi\rangle \equiv \langle\alpha|\hat{x}_\phi|\alpha\rangle = 2\text{Re}(\alpha e^{-i\phi}), \quad \phi = \arg \alpha_L, \quad (2)$$

$$\hat{x}_\phi = \hat{a}e^{-i\phi} + \hat{a}^\dagger e^{i\phi}, \quad (3)$$

where $\hat{x}_\phi$ is the phase-rotated quadrature operator. The rescaled quadrature variable is

$$x \equiv \frac{\mu}{(\eta_1 + \eta_2)CS|\alpha_L|} - \frac{\eta_1 S^2 - \eta_2 C^2}{(\eta_1 + \eta_2)CS}|\alpha_L|, \quad (4)$$

and its variance is

$$\sigma_x \equiv \frac{\eta_1 S^2 + \eta_2 C^2}{[(\eta_1 + \eta_2)CS]^2}. \quad (5)$$

Taking the inverse Fourier transform of the Q symbol (1) gives the antinormally ordered characteristic function and determines the target covariance matrix $\Sigma_m^{(H)}$ [27]. We introduce the reference covariance matrix $\Sigma_{\text{id}}^{(H)}$ by setting $\sigma_N = 0$. With r as an auxiliary squeezing parameter, the two covariance matrices are

$$\Sigma_m^{(H)} = \lim_{r \rightarrow +\infty} R(\phi) \text{diag}(e^{-2r} + \sigma_N, e^{2r}) R^T(\phi), \quad (6)$$

$$\Sigma_{\text{id}}^{(H)} = \lim_{r \rightarrow +\infty} R(\phi) \text{diag}(e^{-2r}, e^{2r}) R^T(\phi), \quad (7)$$

where $\Sigma_{\text{id}}^{(H)}$ is the covariance matrix of the reference homodyne POVM, $R(\phi) = \begin{pmatrix} \cos \phi & -\sin \phi \\ \sin \phi & \cos \phi \end{pmatrix}$ is the rotation matrix, and σ_N is the receiver excess-noise variance,

$$\sigma_N = \sigma_x - 1. \quad (8)$$

For double-homodyne detection, (C_S, S_S) and (C_L, S_L) are the real amplitude coefficients of the signal and LO beam splitters, while (C_i, S_i) are those of homodyne branch $i \in \{1, 2\}$. Each pair obeys $C_j^2 + S_j^2 = 1$ for $j \in \{S, L, 1, 2\}$. The quantities $\eta_k^{(i)}$ and $m_k^{(i)}$ are the quantum efficiency and photon count of detector $k \in \{1, 2\}$ in branch i , and $\mu_i = m_1^{(i)} - m_2^{(i)}$.

The Q symbol of the target double-homodyne POVM is

$$P_G^{(\text{DH})}(x_1, x_2) = \frac{1}{\pi\sqrt{\sigma_1\sigma_2}} \exp\left[-\frac{[x_1 - \text{Re}(\alpha e^{-i\phi})]^2}{\sigma_1} - \frac{[x_2 - \text{Im}(\alpha e^{-i\phi})]^2}{\sigma_2}\right], \quad (9)$$

where $\alpha_L^{(1)} = C_L\alpha_L$ and $\alpha_L^{(2)} = -iS_L\alpha_L$ are the local-oscillator amplitudes in the two homodyne branches, and x_i are the rescaled quadrature variables:

$$x_1 = \frac{1}{2(\eta_1^{(1)} + \eta_2^{(1)})C_1S_1C_S} \left\{ \frac{\mu_1}{|\alpha_L^{(1)}|} - (\eta_1^{(1)}S_1^2 - \eta_2^{(1)}C_1^2)|\alpha_L^{(1)}| \right\}, \quad (10)$$

$$x_2 = \frac{1}{2(\eta_1^{(2)} + \eta_2^{(2)})C_2S_2S_S} \left\{ \frac{\mu_2}{|\alpha_L^{(2)}|} - (\eta_1^{(2)}S_2^2 - \eta_2^{(2)}C_2^2)|\alpha_L^{(2)}| \right\}. \quad (11)$$

The parameters σ_i are defined by

$$\sigma_1 = \frac{\sigma_x^{(1)}}{2C_S^2}, \quad \sigma_2 = \frac{\sigma_x^{(2)}}{2S_S^2}, \quad (12)$$

$$\sigma_x^{(i)} = \frac{\eta_1^{(i)}S_i^2 + \eta_2^{(i)}C_i^2}{[(\eta_1^{(i)} + \eta_2^{(i)})C_iS_i]^2}. \quad (13)$$

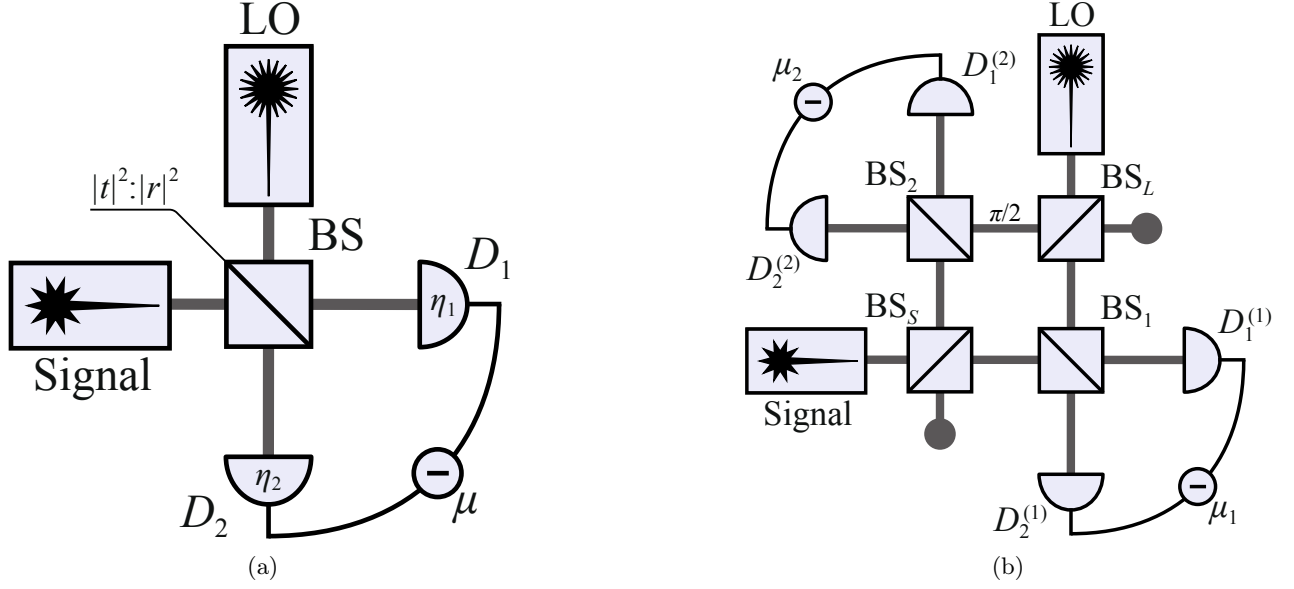


Figure 1: (a) Homodyne receiver. The signal and LO modes enter beam splitter BS with real amplitude coefficients C and S . Photodetectors D_1 and D_2 have quantum efficiencies η_1 and η_2 , and $\mu = m_1 - m_2$ is their photon-count difference. (b) Double-homodyne receiver. Beam splitters BS_S and BS_L split the signal and LO modes, respectively, and the $\pi/2$ phase shifter acts on one LO branch. Each homodyne branch i contains beam splitter BS_i and photodetectors $D_{1,2}^{(i)}$; $\mu_i = m_1^{(i)} - m_2^{(i)}$ is the corresponding photon-count difference.

Thus, for fixed input amplitude α , the conditional outcome variances are $\text{Var}(x_i | \alpha) = \sigma_i/2$.

Following the same procedure used to derive Eq. (6), we obtain the covariance matrix of the target double-homodyne POVM,

$$\Sigma_m^{(\text{DH})} = R(\phi) \text{diag}(\delta_1, \delta_2) R^T(\phi), \quad (14)$$

$$\delta_i = 2\sigma_i - 1. \quad (15)$$

For a complex outcome $z = z_1 + iz_2$, we define the effects of the rank-one reference Gaussian POVM by

$$\hat{\Pi}_{\text{id}}^{(z)} = \frac{1}{\pi} |z, r, \phi\rangle \langle z, r, \phi|, \quad z_1, z_2 \in \mathbb{R}, \quad (16)$$

where

$$|z, r, \phi\rangle = \hat{R}(\phi) \hat{D}(z) \hat{S}(r) |0\rangle = \hat{D}(ze^{i\phi}) \hat{S}(re^{2i\phi}) |0\rangle, \quad (17)$$

and $r \in \mathbb{R}$ is the squeezing parameter. The operators are $\hat{R}(\phi) = e^{i\phi \hat{a}^\dagger \hat{a}}$, $\hat{D}(z) = e^{z\hat{a}^\dagger - z^* \hat{a}}$, and $\hat{S}(\zeta) = e^{[\zeta \hat{a}^{\dagger 2} - \zeta^* \hat{a}^2]/2}$ for $\zeta \in \mathbb{C}$. At $r = 0$, the reference Gaussian POVM reduces to the coherent-state POVM implemented by conventional ideal double-homodyne detection.

For fixed r and ϕ , the displaced squeezed states in Eq. (17) have covariance matrix

$$\Sigma_{\text{id}}^{(\text{DH})} = R(\phi) \begin{pmatrix} e^{2r} & 0 \\ 0 & e^{-2r} \end{pmatrix} R^T(\phi). \quad (18)$$

Complete positivity of the additive-noise channel considered below requires $\Sigma_m^{(\text{DH})} - \Sigma_{\text{id}}^{(\text{DH})} \succeq 0$. Equations (14) and (18) therefore restrict r to the interval fixed by the receiver parameters [19],

$$r \in [r_2, r_1], \quad r_1 = \ln \delta_1^{1/2}, \quad r_2 = \ln \delta_2^{-1/2}. \quad (19)$$

III. GAUSSIAN CHANNEL REPRESENTATIONS OF THE TARGET POVMS

For an arbitrary state $\hat{\rho}$ and POVM density $\hat{\Pi}^{(\mathbf{u})}$, the outcome probability density is [28]

$$p(\mathbf{u}|\hat{\rho}) = \text{Tr} \left[\hat{\rho} \hat{\Pi}^{(\mathbf{u})} \right]. \quad (20)$$

We represent the target POVM by applying a channel $\mathcal{C}_\gamma$ to the state before a reference Gaussian POVM $\hat{\Pi}_{\text{id}}$. Equivalently, the dual map $\mathcal{C}_\gamma^*$ acts on the reference Gaussian POVM [22, 27]:

$$p(\mathbf{u}|\hat{\rho}_G) = \text{Tr} \left[\mathcal{C}_\gamma(\hat{\rho}_G) \hat{\Pi}_{\text{id}}^{(\mathbf{u})} \right] = \text{Tr} \left[\hat{\rho}_G \mathcal{C}_\gamma^* \left(\hat{\Pi}_{\text{id}}^{(\mathbf{u})} \right) \right]. \quad (21)$$

For the channels used here, write $X_{\mathcal{C}}^{(\gamma)} = \kappa_{\mathcal{C}}^{(\gamma)} \mathbb{1}$, where $\kappa_{\Phi}^{(\gamma)} = 1$ and $\kappa_{\mathcal{E}}^{(\gamma)} = \cos \theta_\gamma$. Let $d_{\text{H}} = 1$ and $d_{\text{DH}} = 2$ denote the number of measured quadratures. We rescale the raw reference-POVM outcome as $\mathbf{x} = \mathbf{u}/\kappa_{\mathcal{C}}^{(\gamma)}$. For double-homodyne detection, $\mathbf{u} = (z_1, z_2)^T$ and hence $z = \kappa_{\mathcal{C}}^{(\text{DH})}(x_1 + ix_2)$. The resulting POVM density with respect to $d^{d_\gamma} \mathbf{x}$ is

$$\hat{\Pi}_{\mathcal{C}_\gamma}^{(\mathbf{x})} = \left[\kappa_{\mathcal{C}}^{(\gamma)} \right]^{d_\gamma} \mathcal{C}_\gamma^* \left(\hat{\Pi}_{\text{id}}^{(\kappa_{\mathcal{C}}^{(\gamma)} \mathbf{x})} \right). \quad (22)$$

The prefactor is the Jacobian of the outcome rescaling and preserves the POVM normalization. A channel representation reproduces the target POVM when $\hat{\Pi}_{\mathcal{C}_\gamma}^{(\mathbf{x})} = \hat{\Pi}_m^{(\mathbf{x})}$ for every outcome $\mathbf{x}$.

We restrict $\mathcal{C}_\gamma$ to Gaussian channels. Such a channel is characterized by matrices $X_{\mathcal{C}}^{(\gamma)}$ and $Y_{\mathcal{C}}^{(\gamma)}$. It transforms the first-moment vector $\mathbf{r}_G$ and covariance matrix Σ_G of a Gaussian state as [2]

$$\begin{aligned} \mathbf{r}_G &\mapsto X_{\mathcal{C}}^{(\gamma)} \mathbf{r}_G, \\ \Sigma_G &\mapsto X_{\mathcal{C}}^{(\gamma)} \Sigma_G \left[X_{\mathcal{C}}^{(\gamma)} \right]^T + Y_{\mathcal{C}}^{(\gamma)}. \end{aligned} \quad (23)$$

In the rescaled outcome coordinates, the corresponding covariance-matrix transformation is

$$\Sigma \mapsto \left[\kappa_{\mathcal{C}}^{(\gamma)} \right]^{-2} \left(\Sigma + Y_{\mathcal{C}}^{(\gamma)} \right). \quad (24)$$

We compare additive-noise and attenuator channel representations. At the covariance-matrix level, the matching condition is

$$\Sigma_m^{(\gamma)} = \left[\kappa_{\mathcal{C}}^{(\gamma)} \right]^{-2} \left(\Sigma_{\text{id}}^{(\gamma)} + Y_{\mathcal{C}}^{(\gamma)} \right). \quad (25)$$

Because the channels are centered and $X_{\mathcal{C}}^{(\gamma)}$ is proportional to the identity, the outcome rescaling restores the input-dependent first moments. Equation (25) and the Jacobian in Eq. (22) therefore establish equality with the normalized target Gaussian POVM.

A. Additive-noise channel

First, we consider the additive-noise channel Φ_γ , characterized by the matrices [27]

$$X_{\Phi}^{(\gamma)} = \mathbb{1}, \quad Y_{\Phi}^{(\gamma)} \succeq 0, \quad \gamma \in \{\text{H}, \text{DH}\}. \quad (26)$$

The additive-noise channel is self-dual, $\Phi_\gamma^* = \Phi_\gamma$. It leaves the first moment unchanged and transforms the covariance matrix as

$$\Phi_\gamma : \Sigma \mapsto \Sigma + Y_{\Phi}^{(\gamma)}. \quad (27)$$

Equation (25) requires

$$\Phi_\gamma : \quad \Sigma_{\text{id}}^{(\gamma)} \mapsto \Sigma_m^{(\gamma)} = \Sigma_{\text{id}}^{(\gamma)} + Y_{\Phi}^{(\gamma)}, \quad (28)$$

and therefore

$$Y_{\Phi}^{(\gamma)} = \Sigma_m^{(\gamma)} - \Sigma_{\text{id}}^{(\gamma)} \equiv \Sigma_N^{(\gamma)}, \quad (29)$$

where $\Sigma_N^{(\gamma)}$ is the receiver excess-noise covariance matrix.

For homodyne detection, Eqs. (7) and (6) give

$$\Sigma_N^{(\text{H})} = \Sigma_m^{(\text{H})} - \Sigma_{\text{id}}^{(\text{H})} = R(\phi) \text{diag}(\sigma_N, 0) R^T(\phi) \succeq 0, \quad (30)$$

where σ_N is given by Eq. (8).

For double-homodyne detection, Eqs. (14) and (18) give

$$\Sigma_N^{(\text{DH})}(r) = \Sigma_m^{(\text{DH})} - \Sigma_{\text{id}}^{(\text{DH})} = R(\phi) \begin{pmatrix} \sigma_N^{(1)}(r) & 0 \\ 0 & \sigma_N^{(2)}(r) \end{pmatrix} R^T(\phi) \succeq 0, \quad (31)$$

$$\sigma_N^{(1)}(r) = \delta_1 - e^{2r}, \quad \sigma_N^{(2)}(r) = \delta_2 - e^{-2r}. \quad (32)$$

The squeezing parameter r determines the entries of $\Sigma_N^{(\text{DH})}$. For every $r \in [r_2, r_1]$, the excess-noise covariance matrix is positive semidefinite, $\Sigma_N^{(\text{DH})}(r) \succeq 0$.

B. Attenuator channel

The attenuator channel $\mathcal{E}_\gamma$ is characterized by the matrices [27]

$$X_{\mathcal{E}}^{(\gamma)} = \cos \theta_\gamma \mathbb{1}, \quad Y_{\mathcal{E}}^{(\gamma)} = \sin^2 \theta_\gamma \Sigma_{\text{env}}^{(\gamma)}, \quad \gamma \in \{\text{H}, \text{DH}\}, \quad (33)$$

where $\Sigma_{\text{env}}^{(\gamma)}$ is the covariance matrix of the environmental input state.

Using Eqs. (23) and (25), we choose $\mathcal{E}_\gamma^*$ so that it maps the reference Gaussian POVM to the target POVM. In the rescaled outcome coordinates, its action on the covariance matrix is

$$\mathcal{E}_\gamma^* : \quad \Sigma_{\text{id}}^{(\gamma)} \mapsto \Sigma_m^{(\gamma)} = \frac{1}{\cos^2 \theta_\gamma} \Sigma_{\text{id}}^{(\gamma)} + \tan^2 \theta_\gamma \Sigma_{\text{env}}^{(\gamma)}, \quad \gamma \in \{\text{H}, \text{DH}\}. \quad (34)$$

For homodyne detection, we take the environmental input state to be the vacuum, with covariance matrix

$$\Sigma_{\text{env}}^{(\text{H})} = \mathbb{1}. \quad (35)$$

The attenuator transforms the measured quadrature according to

$$\hat{x}_{\phi, \text{out}} = \cos \theta_{\text{H}} \hat{x}_{\phi, \text{in}} + \sin \theta_{\text{H}} \hat{x}_{\phi, \text{env}}. \quad (36)$$

Let x_{raw} denote the outcome associated with $\hat{x}_{\phi, \text{out}}$. The rescaling $x = x_{\text{raw}} / \cos \theta_{\text{H}}$ corresponds to the rescaled observable

$$\frac{\hat{x}_{\phi, \text{out}}}{\cos \theta_{\text{H}}} = \hat{x}_{\phi, \text{in}} + \tan \theta_{\text{H}} \hat{x}_{\phi, \text{env}}. \quad (37)$$

Since the vacuum quadrature variance is one, the added variance is $\tan^2 \theta_{\text{H}}$. Exact equality with the target homodyne POVM therefore requires

$$\tan^2 \theta_{\text{H}} = \sigma_N. \quad (38)$$

Since $\sigma_x = 1 + \sigma_N$ by Eq. (8), Eq. (38) gives

$$\cos^2 \theta_{\text{H}} = \frac{1}{\sigma_x}, \quad \sin^2 \theta_{\text{H}} = \frac{\sigma_N}{\sigma_x}. \quad (39)$$

For double-homodyne detection, first suppose that $r_2 < r_1$, equivalently $\delta_1 \delta_2 > 1$, and take $r \in (r_2, r_1)$. In the attenuator channel considered here, we take the environmental input state to be a centered pure Gaussian state with covariance matrix

$$\Sigma_{\text{env}}^{(\text{DH})}(r) = R(\phi) \text{diag}(a(r), a(r)^{-1}) R^T(\phi), \quad a(r) > 0. \quad (40)$$

Equations (34), (18), and (14) give

$$\Sigma_m^{(\text{DH})} = R(\phi) \text{diag}(\delta_1, \delta_2) R^T(\phi) = R(\phi) \left\{ \frac{1}{\cos^2 \theta_{\text{DH}}(r)} \text{diag}(e^{2r}, e^{-2r}) + \tan^2 \theta_{\text{DH}}(r) \text{diag}(a(r), a(r)^{-1}) \right\} R^T(\phi), \quad (41)$$

$$\begin{cases} \delta_1 = \frac{e^{2r}}{\cos^2 \theta_{\text{DH}}(r)} + \tan^2 \theta_{\text{DH}}(r) a(r), \\ \delta_2 = \frac{e^{-2r}}{\cos^2 \theta_{\text{DH}}(r)} + \tan^2 \theta_{\text{DH}}(r) a(r)^{-1} \end{cases}. \quad (42)$$

Solving the system yields

$$\cos^2 \theta_{\text{DH}}(r) = \frac{e^{2r} + a(r)}{\delta_1 + a(r)} = \frac{\delta_1 e^{-2r} + \delta_2 e^{2r} - 2}{\delta_1 \delta_2 - 1}, \quad (43)$$

$$a(r) = \frac{\delta_1 - e^{2r}}{\delta_2 e^{2r} - 1}. \quad (44)$$

As $r \rightarrow r_2^+$ or $r \rightarrow r_1^-$, respectively, $a(r)$ or $a(r)^{-1}$ diverges while $\tan^2 \theta_{\text{DH}}(r) \rightarrow 0$. The products $\tan^2 \theta_{\text{DH}}(r) a(r)$ and $\tan^2 \theta_{\text{DH}}(r) a(r)^{-1}$ approach finite limits. The environmental covariance matrix has no finite endpoint limit. The Gaussian channel has the finite limits

$$\lim_{r \rightarrow r_j} X_{\mathcal{E}}^{(\text{DH})}(r) = \mathbb{1}, \quad \lim_{r \rightarrow r_j} Y_{\mathcal{E}}^{(\text{DH})}(r) = \Sigma_N^{(\text{DH})}(r_j). \quad (45)$$

We define $\mathcal{E}_{\text{DH}}(r_j)$ by these one-sided channel limits. This continuous extension equals the corresponding additive-noise channel at each endpoint. If $r_1 = r_2$, then $\delta_1 \delta_2 = 1$, the target POVM is rank one, and both effective channels reduce to the identity channel.

IV. GAUSSIAN-MODULATED COHERENT-STATE PROTOCOL

In the Gaussian-modulated coherent-state (GMCS) protocol, Alice samples the complex amplitude $\alpha = \alpha_1 + i\alpha_2$ from the Gaussian prior:

$$p_A(\alpha) = \frac{1}{2\pi V_A} \exp \left[-\frac{|\alpha|^2}{2V_A} \right]. \quad (46)$$

She then prepares the coherent state $|\alpha\rangle$. The states pass through the phase-insensitive Gaussian communication channel $\mathcal{N}_{A \rightarrow B}$, specified by $X_{\mathcal{N}} = \sqrt{T_{\text{ch}}} \mathbb{1}$ and $Y_{\mathcal{N}} = (1 - T_{\text{ch}} + \xi_{\text{ch}}) \mathbb{1}$, where T_{ch} is the channel transmittance and ξ_{ch} is the output-referred excess-noise variance [29]. The receiver then uses homodyne or double-homodyne detection. The corresponding Alice–Bob mutual information is [19]

$$I_{\text{AB}}^{(\text{H})} = \frac{1}{2} \log_2 \left[1 + \frac{4T_{\text{ch}} V_A}{\sigma_x + \xi_{\text{ch}}} \right], \quad (47)$$

$$I_{\text{AB}}^{(\text{DH})} = \frac{1}{2} \sum_{i=1}^2 \log_2 \left[1 + \frac{4T_{\text{ch}} V_A}{2\sigma_i + \xi_{\text{ch}}} \right], \quad (48)$$

where σ_x and σ_i are given by Eqs. (5) and (12), respectively. The superscripts H and DH label homodyne and double-homodyne detection throughout.

For the covariance-matrix calculations below, we apply the local quadrature transformation $R(\phi)$ on Alice's mode and $R(-\phi)$ on Bob's mode. This transformation preserves the diagonal local blocks and the correlation block proportional to σ_z , while diagonalizing the target and environmental covariance matrices. It does not change the symplectic eigenvalues or the information quantities. All covariance matrices below are expressed in this rotated basis.

The covariance matrix of a two-mode squeezed-vacuum (TMSV) state after Bob's mode passes through the communication channel is [29]

$$\mathcal{I}_A \otimes \mathcal{N}_{A \rightarrow B} : \quad \Sigma_{\text{TMSV}} = \begin{pmatrix} V \mathbb{1} & \sqrt{V^2 - 1} \sigma_z \\ \sqrt{V^2 - 1} \sigma_z & V \mathbb{1} \end{pmatrix} \mapsto \Sigma_{\text{AB}} = \begin{pmatrix} V \mathbb{1} & c \sigma_z \\ c \sigma_z & V_{\text{B}} \mathbb{1} \end{pmatrix}, \quad (49)$$

where $\sigma_z = \text{diag}(1, -1)$, $\mathcal{I}_A$ is the identity channel acting on subsystem A , and the parameters of Σ_{AB} are

$$V = 1 + 4V_A, \quad c = \sqrt{T_{\text{ch}}(V^2 - 1)}, \quad (50)$$

$$V_B = T_{\text{ch}}(V - 1) + 1 + \xi_{\text{ch}}. \quad (51)$$

We next apply either the additive-noise channel Φ_γ (26) or the attenuator channel $\mathcal{E}_\gamma$ (33) to Bob's mode of the TMSV state (49). For each channel, the corresponding reference POVM and outcome rescaling give the mutual information in Eqs. (47) and (48) because they reproduce the same receiver outcome statistics.

It follows that

$$\mathcal{I}_A \otimes \mathcal{E}_\gamma : \Sigma_{AB} \mapsto \Sigma_{AB}^{(\mathcal{E}_\gamma)} = \begin{pmatrix} V\mathbb{1} & c \cos \theta_\gamma \sigma_z \\ c \cos \theta_\gamma \sigma_z & \cos^2 \theta_\gamma (V_B + \tan^2 \theta_\gamma \Sigma_{\text{env}}^{(\gamma)}) \end{pmatrix}, \quad (52)$$

$$\mathcal{I}_A \otimes \Phi_\gamma : \Sigma_{AB} \mapsto \Sigma_{AB}^{(\Phi_\gamma)} = \begin{pmatrix} V\mathbb{1} & c\sigma_z \\ c\sigma_z & V_B\mathbb{1} + \Sigma_N^{(\gamma)} \end{pmatrix}, \quad \gamma \in \{\text{H}, \text{DH}\}. \quad (53)$$

In the untrusted-detector model adopted here, Eve holds a purification of the state $\hat{\rho}_{AB}^{(\mathcal{C}_\gamma)} = (\mathcal{I}_A \otimes \mathcal{C}_\gamma)(\hat{\rho}_{AB})$, and the receiver then applies the corresponding reference Gaussian POVM. The difference between the joint and conditional entropies

$$\chi_{EB}^{(\mathcal{C}_\gamma)} = S_E - S_{E|B} = S_{AB}^{(\mathcal{C}_\gamma)} - S_{A|B}^{(\mathcal{C}_\gamma)}, \quad \mathcal{C}_\gamma \in \{\Phi_\gamma, \mathcal{E}_\gamma\} \quad (54)$$

gives the Holevo information. The joint entropy

$$S_{AB}^{(\mathcal{C}_\gamma)} = g(\nu_1^{(\mathcal{C}_\gamma)}) + g(\nu_2^{(\mathcal{C}_\gamma)}), \quad (55)$$

$$g(\nu) = \frac{\nu + 1}{2} \log_2 \frac{\nu + 1}{2} - \frac{\nu - 1}{2} \log_2 \frac{\nu - 1}{2}$$

is determined by the symplectic eigenvalues of the covariance matrices in Eqs. (52) and (53):

$$\nu_{1,2}^{(\mathcal{C}_\gamma)} = \sqrt{\Delta_{\mathcal{C}_\gamma}/2 \pm \sqrt{\Delta_{\mathcal{C}_\gamma}^2/4 - D_{\mathcal{C}_\gamma}}}, \quad D_{\mathcal{C}_\gamma} = \det \Sigma_{AB}^{(\mathcal{C}_\gamma)}, \quad (56)$$

$$\Delta_{\mathcal{C}_\gamma} = \det([\Sigma_{AB}^{(\mathcal{C}_\gamma)}]_{11}) + \det([\Sigma_{AB}^{(\mathcal{C}_\gamma)}]_{22}) + 2 \det([\Sigma_{AB}^{(\mathcal{C}_\gamma)}]_{12}), \quad (57)$$

where $[\Sigma_{AB}^{(\mathcal{C}_\gamma)}]_{ii}$ ($[\Sigma_{AB}^{(\mathcal{C}_\gamma)}]_{ij}$) are diagonal (off-diagonal) blocks of $\Sigma_{AB}^{(\mathcal{C}_\gamma)}$.

The conditional entropies $S_{E|B} = S_{A|B}^{(\gamma)}$ are calculated from the corresponding conditional covariance matrices,

$$S_{A|B}^{(\gamma)} = g(\nu_3^{(\gamma)}), \quad \nu_3^{(\gamma)} = \sqrt{\det \Sigma_{A|B}^{(\gamma)}}, \quad (58)$$

$$\Sigma_{A|B}^{(\gamma)} = V\mathbb{1} - c^2 \sigma_z (V_B\mathbb{1} + \Sigma_m^{(\gamma)})^{-1} \sigma_z. \quad (59)$$

Direct computations show that these conditional matrices, and therefore their symplectic eigenvalues, are independent of which effective receiver channel is used. The symplectic eigenvalues are

$$\nu_3^{(\text{H})} = \sqrt{V \left(V - \frac{c^2}{V_B + \sigma_N} \right)}, \quad (60)$$

$$\nu_3^{(\text{DH})} = V \sqrt{\frac{(V_B + \delta_1 - \frac{c^2}{V})(V_B + \delta_2 - \frac{c^2}{V})}{(V_B + \delta_1)(V_B + \delta_2)}}, \quad (61)$$

The asymptotic secret fraction is given by the Devetak–Winter formula [30]

$$K^{(\mathcal{C}_\gamma)} = \beta I_{AB}^{(\gamma)} - \chi_{EB}^{(\mathcal{C}_\gamma)}, \quad \gamma \in \{\text{H}, \text{DH}\}, \mathcal{C}_\gamma \in \{\Phi_\gamma, \mathcal{E}_\gamma\}. \quad (62)$$

Here $\beta \in (0, 1]$ is the reverse-reconciliation efficiency.

The additive-noise symplectic eigenvalues obtained from Eq. (56) do not simplify further. The following sections give analytic expressions only for the attenuator channel.

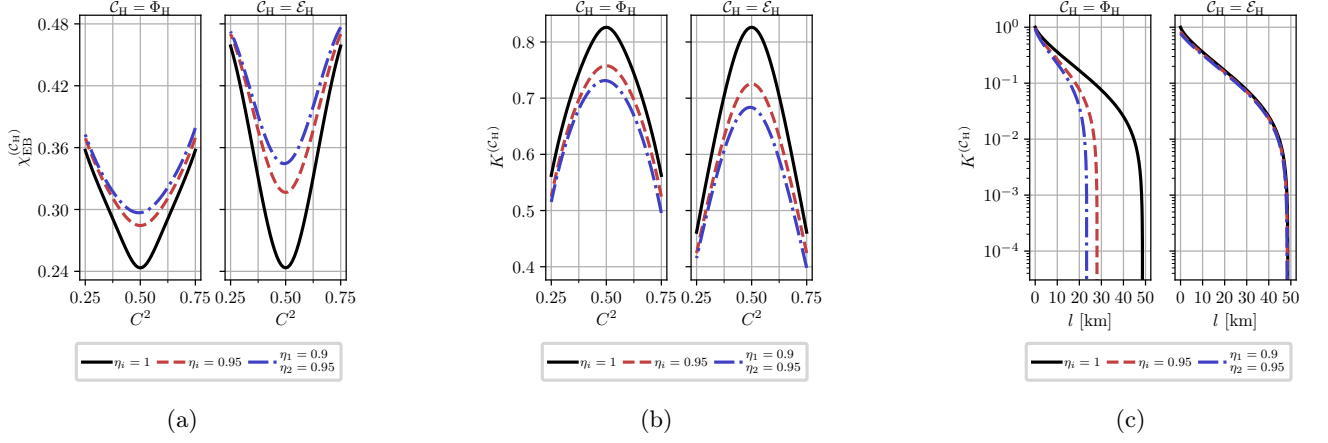


Figure 2: (a) Holevo information, $\chi_{\text{EB}}^{(\mathcal{C}_H)}$ and (b) asymptotic secret fraction, $K^{(\mathcal{C}_H)}$, as a function of the beam-splitter transmittance C^2 of the homodyne receiver, computed at different values of the detector efficiencies for the additive-noise and attenuator channels, as indicated in the panel title. The parameters are: $V = 5$; $T_{\text{ch}} = 0.95$; $\xi_{\text{ch}} = 10^{-2}$; and $\beta = 0.95$. (c) $K^{(\mathcal{C}_H)}$ versus channel length l for the additive-noise and attenuator channels. The parameters are: $V = 5$; $\beta = 0.95$; $\xi_{\text{ch}} = 10^{-2}$; losses are 20 dB per 100 km.

A. Homodyne detection

For the attenuator channel, Eqs. (52) and (39) give

$$\Sigma_{\text{AB}}^{(\mathcal{E}_H)} = \begin{pmatrix} V\mathbb{1} & \frac{c}{\sqrt{\sigma_x}}\sigma_z \\ \frac{c}{\sqrt{\sigma_x}}\sigma_z & \frac{1}{\sigma_x}(V_{\text{B}} + \sigma_N)\mathbb{1} \end{pmatrix} \equiv \begin{pmatrix} V\mathbb{1} & \tilde{c}\sigma_z \\ \tilde{c}\sigma_z & \tilde{V}_{\text{B}}\mathbb{1} \end{pmatrix}, \quad (63)$$

$$\tilde{V}_{\text{B}} \equiv T(V - 1) + 1 + \xi, \quad (64)$$

$$\tilde{c} \equiv \sqrt{T(V^2 - 1)}, \quad (65)$$

where $\tilde{V}_{\text{B}}$ is Bob's scaled variance and $\tilde{c}$ is the scaled Alice–Bob correlation amplitude. The effective transmittance T and excess-noise variance ξ are

$$T = T_{\text{ch}} \cdot \frac{1}{\sigma_x}, \quad (66a)$$

$$\xi \equiv \xi_{\text{ch}} \cdot \frac{1}{\sigma_x}. \quad (66b)$$

For homodyne detection, receiver asymmetry therefore scales both the effective transmittance (66a) and the output-referred excess-noise variance (66b) by σ_x^{-1} .

The symplectic eigenvalues of $\Sigma_{\text{AB}}^{(\mathcal{E}_H)}$ are [2]

$$\nu_{1,2}^{(\mathcal{E}_H)} = \frac{\omega_{\text{H}} \pm [\tilde{V}_{\text{B}} - V]}{2}, \quad \omega_{\text{H}} = \sqrt{(V + \tilde{V}_{\text{B}})^2 - 4\tilde{c}^2}. \quad (67)$$

These eigenvalues are then substituted into Eq. (55) to obtain the joint entropy $S_{\text{AB}}^{(\mathcal{E}_H)}$.

Figure 2 shows the Eve–Bob Holevo information and asymptotic secret fraction as functions of the receiver beam-splitter transmittance C^2 , together with the asymptotic secret fraction as a function of channel length, for both channels. The attenuator scales the effective transmittance and output-referred excess-noise variance by the same factor (66). For the parameters shown, the attenuator channel gives higher Holevo information and a lower asymptotic secret fraction at $T_{\text{ch}} = 0.95$. In panel (c), its asymptotic secret fraction remains positive up to a greater channel length than for the additive-noise channel.

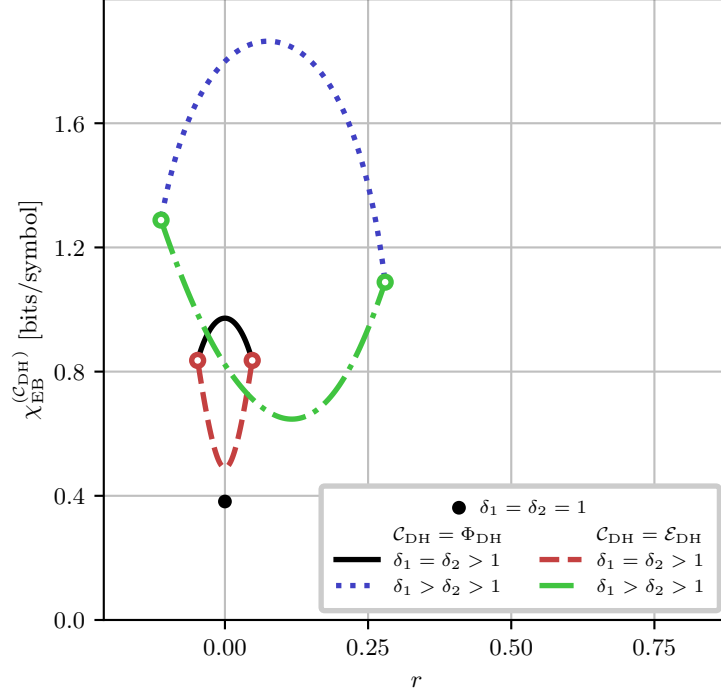


Figure 3: Holevo information, $\chi_{\text{EB}}^{(\mathcal{C}_{\text{DH}})}$, as a function of the squeezing parameter r at different values of $\delta_{1,2}$. The black dot marks ideal detection. Solid and dash-dotted curves show the additive-noise channel; dashed and dotted curves show the attenuator channel. Color-matched white-filled dots mark the finite endpoint limits of the attenuator curves. At each endpoint, the limiting channel equals the corresponding additive-noise channel; the environmental covariance matrix has no finite limit there (see the discussion below Eq. (44)). Values of $\delta_{1,2}$ are: $\delta_{1,2} = 1.1$ for solid and dashed curves, $\delta_1 = 1.75, \delta_2 = 1.25$ for dash-dotted and dotted curves. The channel parameters are: $V = 5$; $T_{\text{ch}} = 0.95$; and $\xi_{\text{ch}} = 10^{-2}$.

B. Double-homodyne detection

For the attenuator channel, Eqs. (52) and (43) give

$$\Sigma_{\text{AB}}^{(\mathcal{E}_{\text{DH}})} = \begin{pmatrix} V\mathbb{I} & c \cos \theta_{\text{DH}}(r) \sigma_z & 0 \\ c \cos \theta_{\text{DH}}(r) \sigma_z & \cos^2 \theta_{\text{DH}}(r) \left(V_{\text{B}} + \delta_1 - \frac{e^{2r}}{\cos^2 \theta_{\text{DH}}(r)} \right) & 0 \\ 0 & 0 & \cos^2 \theta_{\text{DH}}(r) \left(V_{\text{B}} + \delta_2 - \frac{e^{-2r}}{\cos^2 \theta_{\text{DH}}(r)} \right) \end{pmatrix}. \quad (68)$$

The symplectic eigenvalues of this matrix can be calculated using invariants as in Eq. (56). These eigenvalues, and hence the joint entropy, depend explicitly on the squeezing parameter r , as in the additive-noise channel [19].

Figure 3 shows the Holevo information as a function of r for both channels. For ideal double-homodyne detection, $\delta_1 = \delta_2 = 1$ and the admissible interval collapses to $r = 0$ [19]. Both effective channels then reduce to the identity channel and give the same Holevo information, shown by the black dot.

When $\delta_1 = \delta_2 > 1$, the Holevo information is symmetric about $r = 0$ for both channels. It is maximal at $r = 0$ for the additive-noise channel and minimal there for the attenuator channel. In the continuous extension defined below Eq. (44), the attenuator channel equals the corresponding additive-noise channel at $r = r_2$ and $r = r_1$. The Holevo information therefore agrees between the two channels at each endpoint, as shown by the color-matched white-filled dots in Fig. 3.

For the asymmetric parameter sets considered, where $\delta_1 \neq \delta_2$, neither curve is symmetric in r , and the joint entropies at the two endpoints $r = r_2$ and $r = r_1$ are unequal. For these parameter sets, the Holevo information is concave in r for the additive-noise channel and convex in r for the attenuator channel.

Figure 4 shows the ranges of the asymptotic secret fraction obtained by varying r over $[r_2, r_1]$ for both channels. For the parameter sets shown, convexity places the maximum of the continuously extended attenuator Holevo information

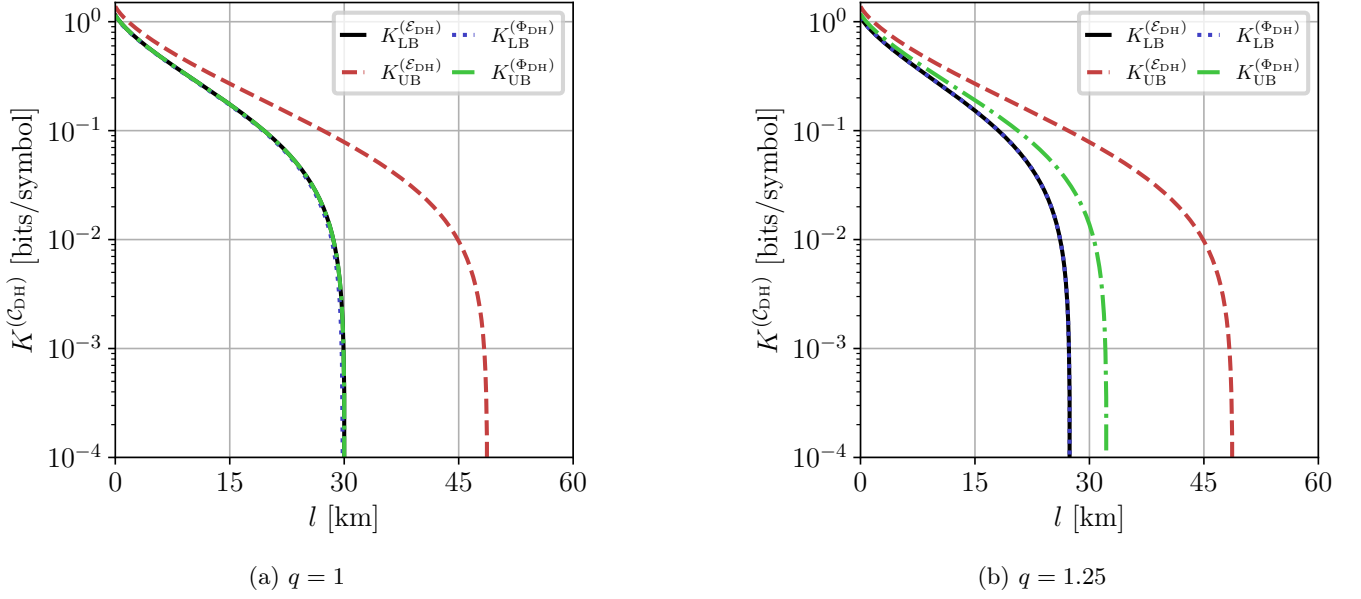


Figure 4: Asymptotic secret fraction $K^{(C_{DH})}$ versus channel length l , computed for fixed $\sigma_x^{(i)} = 1.05$ (see Eq. (13)) and various imbalance ratios $q \equiv \frac{S_S^2}{C_S^2}$ for the additive-noise and attenuator channels (see Fig. 3). The parameters are: $V = 5$; $\beta = 0.95$; $\xi_{ch} = 10^{-2}$; the fiber loss is 20 dB per 100 km.

at an endpoint, where the limiting channel equals the corresponding additive-noise channel. For the near-ideal value $\sigma_x^{(i)} = 1.05$, the additive-noise secret-fraction range is narrow, so the lower bounds for the two channels nearly coincide. In Fig. 4a, the channel lengths at which the upper and lower additive-noise bounds vanish differ by only 0.2 km. For both values of q shown, the attenuator range is wider than the additive-noise range. Increasing q from 1 to 1.25 makes the additive-noise range more pronounced.

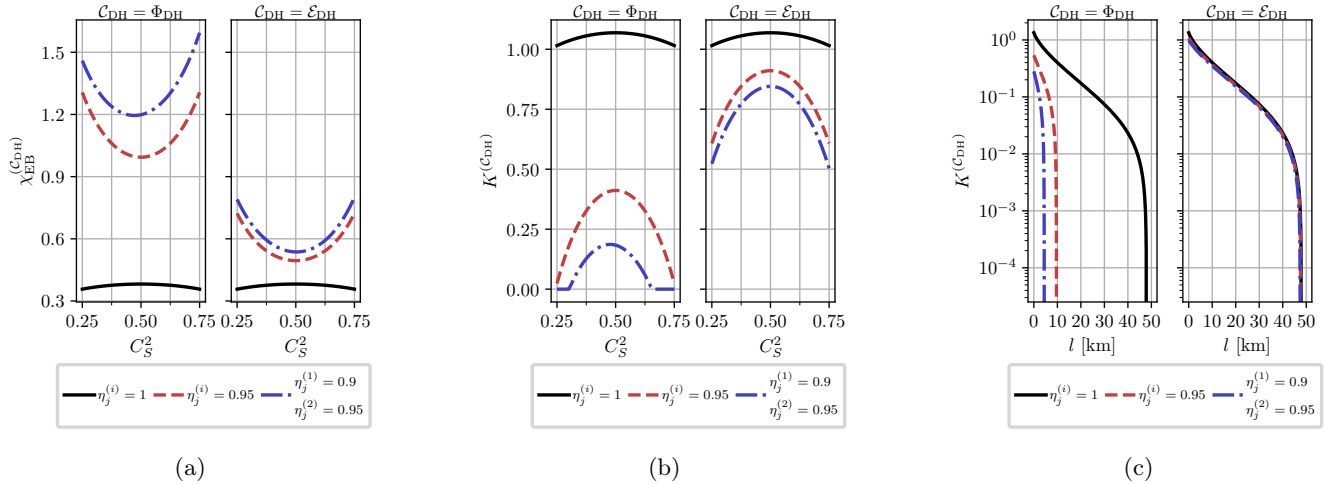


Figure 5: (a) Holevo information, $\chi_{EB}^{(C_{DH})}$ and (b) asymptotic secret fraction, $K^{(C_{DH})}$, as a function of the signal beam-splitter transmittance C_S^2 of the double-homodyne receiver, computed at different values of the detector efficiencies for the additive-noise and attenuator channels, as indicated in the panel title. The parameters are: $V = 5$; $T_{ch} = 0.95$; $\xi_{ch} = 10^{-2}$; and $\beta = 0.95$. (c) $K^{(C_{DH})}$ versus channel length l for the additive-noise and attenuator channels. The parameters are: $V = 5$; $\beta = 0.95$; $\xi_{ch} = 10^{-2}$; losses are 20 dB per 100 km. For both channels $r = \frac{r_2 + r_1}{2}$ (see main text).

Figure 5 shows the Eve–Bob Holevo information and asymptotic secret fraction as functions of the signal beam-

splitter transmittance C_S^2 , together with the asymptotic secret fraction as a function of channel length, for both channels. For $\eta_{1,2}^{(i)} = 1$, $i = 1, 2$, the target POVM is rank one for every C_S^2 ; both effective channels reduce to the identity channel and give equal joint entropies, shown by the black curve. For the nonunit detector efficiencies shown, the attenuator channel gives lower Holevo information than the additive-noise channel. The separation reflects both the choice of effective channel and the fixed value $r = (r_1 + r_2)/2$. For symmetric measurement noise, the midpoint of the admissible interval is $r = 0$. This value maximizes the additive-noise Holevo information and minimizes the attenuator Holevo information. For the parameters in panel (c), the attenuator asymptotic secret fraction remains positive up to a greater channel length.

V. DISCUSSION AND CONCLUSION

In this paper, we have constructed and compared additive-noise and attenuator channel representations of the target homodyne and double-homodyne POVMs. Our approach requires the dual map of the selected Gaussian channel, together with a deterministic invertible rescaling of the measurement outcomes, to reproduce the target POVM. At the level of covariance matrices, this condition is expressed by Eq. (25). It constrains the channel parameters, which are then determined by the receiver parameters and, for double-homodyne detection, by the squeezing parameter r .

After the deterministic invertible rescaling of the measurement outcomes, both channel representations reproduce the same conditional statistics of Bob's outcomes. They therefore give the same Alice-Bob mutual information. For fixed source and channel parameters, the conditional covariance matrix in Eq. (59), and hence the conditional entropy $S_{A|B}^{(\gamma)}$, are the same for both channels because the detector imperfections enter the conditional state through the same target POVM. The difference between the security estimates originates from the joint state of Alice and Bob. The additive-noise and attenuator channels generally produce different joint covariance matrices and, consequently, different joint entropies. In the untrusted-detector model considered here, Eve holds a purification of the Alice-Bob state after the effective receiver channel. The corresponding Holevo information (54) can therefore depend on the selected channel while Bob's observed statistics remain unchanged.

For homodyne detection, we take the environmental mode of the attenuator channel to be in the vacuum state, so that the detector imperfections are represented by a pure-loss channel. This channel rescales both the effective transmission and the output-referred excess noise by the same factor, as shown in Eq. (66).

The numerical results in Fig. 2 show that the additive-noise and attenuator channels give different security estimates. At high channel transmittance and for the parameters considered, the attenuator channel gives a larger Holevo information and a lower asymptotic secret fraction. Its secret fraction remains positive over a greater channel length. Thus, the channel giving the lower secret fraction changes with the communication-channel transmittance.

A distinguishing feature of double-homodyne detection is the nonuniqueness of the channel representation of its target POVM. The covariance matrix of the target double-homodyne POVM is fixed by the receiver parameters and does not depend on the squeezing parameter r . The effects of the reference Gaussian POVM are proportional to projectors onto displaced squeezed states and depend on r . The target POVM is obtained by applying the dual map of the selected Gaussian channel to the reference Gaussian POVM, together with the outcome rescaling. The requirement that the excess-noise covariance matrix be positive semidefinite restricts r to the interval given by Eq. (19).

For the additive-noise channel, variation of r redistributes the measurement noise between the two conjugate quadratures according to Eq. (32). For the attenuator channel, the same variation changes both the attenuation coefficient and the covariance matrix of the environmental state, as described by Eqs. (43) and (44). In the attenuator channel considered here, the generally anisotropic measurement noise is reproduced by an environmental mode in an anisotropic Gaussian state. We choose this environmental state to be pure.

When the admissible interval for r has nonzero width, one additive-noise quadrature variance vanishes at each boundary. The environmental state in the attenuator channel then approaches an infinitely squeezed limit, with one quadrature variance tending to zero and the conjugate variance diverging. At each boundary, the attenuator channel converges to the corresponding additive-noise channel. If the interval collapses to a single value, the target POVM is rank one and both effective channels reduce to the identity channel.

Figure 3 shows that the Holevo information depends on r in qualitatively different ways for the two channels. For the parameter sets considered, this dependence is concave for the additive-noise channel and convex for the attenuator channel. When the measurement noise is symmetric between the two quadratures, both curves are symmetric about $r = 0$. In this case, the coherent-state POVM at $r = 0$ maximizes the Holevo information for the additive-noise channel and minimizes it for the attenuator channel. For the asymmetric case considered, this symmetry is lost and the extrema are shifted away from $r = 0$.

The lower and upper secret-fraction curves in Fig. 4 describe the ranges obtained by varying r over its admissible interval for the two channels. For the near-ideal quadrature variances used in Fig. 4, the additive-noise range is

narrow. The lower bound of the continuously extended attenuator curve occurs at an endpoint, where the limiting channel equals the corresponding additive-noise channel. These facts make the lower bounds nearly coincide. For both imbalance ratios shown, the attenuator range is wider than the additive-noise range. Increasing the imbalance from $q = 1$ to $q = 1.25$ makes the additive-noise range more pronounced.

The comparison shown in Fig. 5 was performed at the midpoint of the admissible interval of r . In the symmetric case, this value coincides with the maximum of the additive-noise Holevo information and the minimum of the attenuator Holevo information. The pronounced separation between the secret fractions in Fig. 5 therefore reflects both the choice of effective channel and the selected squeezing parameter.

These results clarify the role of the effective receiver channel when receiver noise is treated as untrusted. The additive-noise and attenuator channels reproduce the same receiver statistics after outcome rescaling. They generally yield different joint states of Alice and Bob. In the untrusted-detector model, Eve holds a purification of the state produced by the selected channel. Selecting that channel therefore fixes an assumption in the security analysis. The effective receiver channel should follow from the physical structure of the receiver. If several channel representations remain compatible with the receiver statistics, the analysis should report the dependence of the security estimate on that choice. The present comparison covers the additive-noise channels defined above and attenuator channels with pure Gaussian environmental input states for $r \in (r_2, r_1)$, together with their endpoint limits.

Appendix A extends the receiver model to include independent pure losses at the output ports of a lossless beam splitter. These losses can be incorporated into the effective photodetector efficiencies. For a general lossy asymmetric beam splitter, the positive factor in the polar decomposition need not be diagonal in the output basis and therefore cannot be absorbed into independent detector-efficiency replacements.

The security analysis treats the GMCS CV-QKD protocol in the asymptotic regime under collective Gaussian attacks and the untrusted-detector model. Under these assumptions, identical receiver statistics can lead to different estimates of Eve's information for different effective receiver channels.

ACKNOWLEDGEMENTS

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Appendix A: Lossy beam splitter

In this appendix, we show that independent pure-loss channels at the output ports of a lossless beam splitter are equivalent to a single lossy beam splitter.

First, consider the case of homodyne detection. The lossless unbalanced beam splitter mixes the input signal mode $\hat{a}$ and the local oscillator mode $\hat{a}_L$. This transformation is described by the unitary rotation matrix $\mathbf{U}_{\text{BS}}$,

$$\begin{aligned} \begin{pmatrix} \hat{a}_1 \\ \hat{a}_2 \end{pmatrix} &= \mathbf{U}_{\text{BS}} \begin{pmatrix} \hat{a} \\ \hat{a}_L \end{pmatrix} = \begin{pmatrix} C & S \\ -S & C \end{pmatrix} \begin{pmatrix} \hat{a} \\ \hat{a}_L \end{pmatrix} \\ &= \begin{pmatrix} C\hat{a} + S\hat{a}_L \\ -S\hat{a} + C\hat{a}_L \end{pmatrix}, \end{aligned} \quad (\text{A1})$$

where C and S are the amplitude transmission and reflection coefficients of the beam splitter BS, respectively.

We model the subsequent losses by passing each intermediate output mode $\hat{a}_{1,2}$ through a virtual beam splitter of transmittance $\tau_{1,2}$. This mixes each signal with an independent environmental vacuum mode $\hat{v}_{1,2}$, where $[\hat{v}_i, \hat{v}_j^\dagger] = \delta_{ij}$,

$$\begin{pmatrix} \hat{b}_1 \\ \hat{c}_1 \end{pmatrix} = \begin{pmatrix} \sqrt{\tau_1} & -\sqrt{1-\tau_1} \\ \sqrt{1-\tau_1} & \sqrt{\tau_1} \end{pmatrix} \begin{pmatrix} \hat{a}_1 \\ \hat{v}_1 \end{pmatrix}, \quad (\text{A2})$$

$$\begin{pmatrix} \hat{b}_2 \\ \hat{c}_2 \end{pmatrix} = \begin{pmatrix} \sqrt{\tau_2} & -\sqrt{1-\tau_2} \\ \sqrt{1-\tau_2} & \sqrt{\tau_2} \end{pmatrix} \begin{pmatrix} \hat{a}_2 \\ \hat{v}_2 \end{pmatrix}. \quad (\text{A3})$$

Tracing out the lost environmental modes $\hat{c}_{1,2}$, the remaining physical output modes $\hat{b}_{1,2}$ expand to

$$\begin{pmatrix} \hat{b}_1 \\ \hat{b}_2 \end{pmatrix} = \begin{pmatrix} \sqrt{\tau_1}C & \sqrt{\tau_1}S \\ -\sqrt{\tau_2}S & \sqrt{\tau_2}C \end{pmatrix} \begin{pmatrix} \hat{a} \\ \hat{a}_L \end{pmatrix} + \begin{pmatrix} -\sqrt{1-\tau_1}\hat{v}_1 \\ -\sqrt{1-\tau_2}\hat{v}_2 \end{pmatrix}. \quad (\text{A4})$$

From Eq. (A4), the effective scattering matrix is

$$\mathbf{T}_{\text{BS}} = \begin{pmatrix} \sqrt{\tau_1}C & \sqrt{\tau_1}S \\ -\sqrt{\tau_2}S & \sqrt{\tau_2}C \end{pmatrix}. \quad (\text{A5})$$

The associated noise operators are $\hat{F}_i = -\sqrt{1-\tau_i} \hat{v}_i$ for $i \in \{1, 2\}$.

Preservation of the canonical commutation relations $[\hat{b}_i, \hat{b}_j^\dagger] = \delta_{ij}$ for the output fields requires

$$[\hat{F}_i, \hat{F}_j^\dagger] = \delta_{ij} - \sum_k T_{ik} T_{jk}^*, \quad (\text{A6})$$

where T_{ik} are the elements of the scattering matrix $\mathbf{T}_{\text{BS}}$ (A5). Evaluating this constraint explicitly using the elements of $\mathbf{T}_{\text{BS}}$ yields:

$$[\hat{F}_1, \hat{F}_1^\dagger] = 1 - \tau_1, \quad (\text{A7})$$

$$[\hat{F}_2, \hat{F}_2^\dagger] = 1 - \tau_2, \quad (\text{A8})$$

$$[\hat{F}_1, \hat{F}_2^\dagger] = 0, \quad (\text{A9})$$

which matches the commutators calculated from Eq. (A4).

We now relate this setup to the standard input-output description of lossy beam splitters [31, 32]. This description uses an interaction matrix $\mathbf{A}$ acting on ancillary modes and requires $\mathbf{1} - \mathbf{A}\mathbf{A}^\dagger \succeq 0$. The scattering matrix $\mathbf{T}_{\text{BS}}$ in Eq. (A5) factors into a diagonal attenuation matrix and the lossless unitary:

$$\begin{aligned} \mathbf{T}_{\text{BS}} &= \begin{pmatrix} \sqrt{\tau_1} & 0 \\ 0 & \sqrt{\tau_2} \end{pmatrix} \begin{pmatrix} C & S \\ -S & C \end{pmatrix} \\ &= \begin{pmatrix} \sqrt{\tau_1} & 0 \\ 0 & \sqrt{\tau_2} \end{pmatrix} \mathbf{U}_{\text{BS}}. \end{aligned} \quad (\text{A10})$$

For $\mathbf{A}\mathbf{A}^\dagger = \text{diag}(1 - \tau_1, 1 - \tau_2)$,

$$\sqrt{\mathbf{1} - \mathbf{A}\mathbf{A}^\dagger} = \begin{pmatrix} \sqrt{\tau_1} & 0 \\ 0 & \sqrt{\tau_2} \end{pmatrix}. \quad (\text{A11})$$

Equation (A10) then gives the polar decomposition [32]

$$\mathbf{T}_{\text{BS}} = \sqrt{\mathbf{1} - \mathbf{A}\mathbf{A}^\dagger} \mathbf{U}_{\text{BS}}. \quad (\text{A12})$$

Because $\mathbf{A}\mathbf{A}^\dagger$ is diagonal, the two output losses can be absorbed into the detector efficiencies as $\eta_{1,2} \rightarrow \tau_{1,2}\eta_{1,2}$. The same rescaling applies to double-homodyne detection.

We finally consider a general lossy asymmetric beam splitter with scattering matrix [33]

$$\mathbf{S} = \begin{pmatrix} t & \rho \\ r_{\text{BS}} & \tau \end{pmatrix}, \quad \mathbf{S}\mathbf{S}^\dagger \leq \mathbf{1}, \quad (\text{A13})$$

where t and τ are complex transmission amplitudes, while ρ and r_{BS} are complex reflection amplitudes. A general $\mathbf{S}$ also admits a polar decomposition. Its positive factor need not be diagonal in the output basis, so it cannot be absorbed into two detector-efficiency replacements. The same obstruction appears in the independent interference coefficients in σ_G and μ_G below:

$$|\alpha, \alpha_L\rangle \mapsto |\alpha_{\text{out},1}, \alpha_{\text{out},2}\rangle, \quad (\text{A14})$$

$$\alpha_{\text{out},1} = t\alpha + \rho\alpha_L, \quad \alpha_{\text{out},2} = r_{\text{BS}}\alpha + \tau\alpha_L, \quad (\text{A15})$$

We use the normalized Gaussian density

$$G(x; \sigma) \equiv \frac{1}{\sqrt{2\pi}\sigma} \exp\left(-\frac{x^2}{2\sigma}\right). \quad (\text{A16})$$

Then

$$P_G(\mu) = G(\mu - \mu_G; \sigma_G), \quad (\text{A17})$$

$$\begin{aligned} \sigma_G &= \eta_1 |\alpha_{\text{out},1}|^2 + \eta_2 |\alpha_{\text{out},2}|^2 \\ &= (\eta_1 |\rho|^2 + \eta_2 |\tau|^2) |\alpha_L|^2 + (\eta_1 |t|^2 + \eta_2 |r_{\text{BS}}|^2) |\alpha|^2 \\ &\quad + 2 \text{Re}[(\eta_1 t \rho^* + \eta_2 r_{\text{BS}} \tau^*) \alpha \alpha_L^*] \\ &\approx (\eta_1 |\rho|^2 + \eta_2 |\tau|^2) |\alpha_L|^2 + 2 \text{Re}[(\eta_1 t \rho^* + \eta_2 r_{\text{BS}} \tau^*) \alpha \alpha_L^*], \end{aligned} \quad (\text{A18})$$

$$\begin{aligned} \mu_G &= \eta_1 |\alpha_{\text{out},1}|^2 - \eta_2 |\alpha_{\text{out},2}|^2 \\ &= (\eta_1 |\rho|^2 - \eta_2 |\tau|^2) |\alpha_L|^2 + (\eta_1 |t|^2 - \eta_2 |r_{\text{BS}}|^2) |\alpha|^2 \\ &\quad + 2 \text{Re}[(\eta_1 t \rho^* - \eta_2 r_{\text{BS}} \tau^*) \alpha \alpha_L^*] \\ &\approx (\eta_1 |\rho|^2 - \eta_2 |\tau|^2) |\alpha_L|^2 + 2 \text{Re}[(\eta_1 t \rho^* - \eta_2 r_{\text{BS}} \tau^*) \alpha \alpha_L^*]. \end{aligned} \quad (\text{A19})$$

Appendix B: Fluctuating local oscillator

In this appendix, we extend the homodyne model to include classical amplitude fluctuations of a phase-locked local oscillator (LO).

Let $L = |\alpha_L|$ denote the mean LO amplitude. For fixed L , the photon-count-difference distribution in the strong-LO Gaussian approximation is [19]

$$P_G(\mu | L) = G(\mu - \mu_G(L); \sigma_G(L)), \quad (\text{B1})$$

$$\sigma_G(L) \approx (\eta_1 S^2 + \eta_2 C^2) L^2, \quad (\text{B2})$$

$$\mu_G(L) \approx (\eta_1 S^2 - \eta_2 C^2) L^2 + CS(\eta_1 + \eta_2) L \langle \hat{x}_\phi \rangle. \quad (\text{B3})$$

Here G is defined in Eq. (A16).

We write the instantaneous LO amplitude as $L + \Delta_L$, where Δ_L is a zero-mean Gaussian random variable:

$$p_L(\Delta_L) = G(\Delta_L; \sigma_L), \quad \sqrt{\sigma_L} \ll L. \quad (\text{B4})$$

Here σ_L is the variance of the LO amplitude fluctuations. Under the small-fluctuation condition, extending the integral over Δ_L to the full real line introduces a negligible tail with $L + \Delta_L < 0$.

The unconditional photon-count-difference distribution is the Gaussian mixture:

$$P_{\text{tot}}(\mu) = \int_{-\infty}^{\infty} d\Delta_L G(\Delta_L; \sigma_L) P_G(\mu | L + \Delta_L). \quad (\text{B5})$$

We use a linearized Gaussian approximation: we set $\sigma_G(L + \Delta_L) \approx \sigma_G(L)$ and linearize the conditional mean,

$$\mu_G(L + \Delta_L) \approx \mu_G(L) + \mu'_G(L) \Delta_L, \quad (\text{B6})$$

$$\mu'_G(L) = 2(\eta_1 S^2 - \eta_2 C^2) L + CS(\eta_1 + \eta_2) \langle \hat{x}_\phi \rangle. \quad (\text{B7})$$

This approximation retains the broadening generated by the linearized conditional mean, freezes the conditional variance at $\sigma_G(L)$, and discards the quadratic term in $\mu_G(L + \Delta_L)$, including its contribution to the mean. The Gaussian integral then gives

$$P_{\text{tot}}(\mu) \approx G(\mu - \mu_G(L); \sigma_G(L) + [\mu'_G(L)]^2 \sigma_L). \quad (\text{B8})$$

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